CSE4203: Principles of Software Architecture

The Pharaohs' Twinverse

Milestone 1: Proposal & Architectural Requirements

Project	The Pharaohs' Twinverse
Team name	The Pharaohs
Milestone	M1: Proposal & architectural requirements
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Contribution table

Member	Contribution
Youssef Mohamed	Section 1 (System overview), Section 3 (Context diagram)
Moustafa Waleed	Section 2 (Stakeholders), Section 6 (Constraints)
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Baraa Ayman	Section 5 (Quality attribute scenarios)
Fatma Khalil	Section 7 (ASRs and utility tree), references

Proposal

Project idea

The Pharaohs' Twinverse is a heritage-conservation platform that helps conservators and museum teams assess damage on statues and other heritage objects. Field staff upload photos, videos or scans with provenance details; the system checks media quality, detects and segments damage (cracks first) and calculates basic measurements. Conservators confirm the object and material, correct the marked damage and approve every report at three human review gates, so no AI output becomes a record without expert sign-off. Reports are built from verified conservation sources and cite them; when the evidence is weak, the system requests expert examination instead of recommending a treatment. All artifact records, annotations and decisions are versioned and audited. Image restoration, 3D reconstruction and the talking statue are future extensions.

How the idea meets the five approval requirements

#	Requirements	How The Pharaohs' Twinverse meets it
1	**User roles**	Five primary roles: **field technician / photographer** (uploads media), **conservator** (confirms, corrects and approves), **museum curator / collection manager** (artifact records and reports), **platform administrator** (users, permissions, audit logs) and **ML engineer / annotator** (evaluation and annotation). Researchers and museum visitors are future roles. *(Sections 2 and 4)*
2	**Competing attributes**	**Performance vs trustworthiness:** evidence checks add response time, but reports must stay trustworthy. **Security and integrity vs usability:** review gates and protected records add steps for users. **Modifiability vs performance:** flexible analysis capabilities must still deliver timely results. *(Section 7.4; QAS-1, 3, 4, 5, 7)*
3	**External integration**	**Heritage reference sources** (verified, licence-checked conservation manuals and papers) ground the cited reports, and **camera / 3D scanner** devices supply artifact images and scans. Planned: the institution's collection system (FR-12) and an external speech service for the talking statue (future). *(Section 3 context diagram)*
4	**Separated parts**	Core responsibilities are kept separate: media intake, damage analysis, review, reports, and records with access control. Damage analysis is deliberately isolated from review and reporting (ASR-1) so it can change on its own, and the user-facing interface is separate from the analysis, evidence and record-keeping parts. *(Sections 1 and 7.3)*
5	**Realistic scope**	The prototype covers media intake and quality checks, damage analysis (crack detection and segmentation), basic measurements, expert review, cited reports, versioned records and access control: nine user stories (FR-01 to FR-05 and FR-07 to FR-10). Multi-class damage, image restoration, 3D reconstruction and the talking statue stay out of scope. *(Sections 1, 4 and 6)*

1. System overview

Problem. Conservators and museum teams often assess damage on statues and heritage objects by visual inspection and manual records. Assessments can be slow, subjective, and difficult to compare over time, while evidence and review history may be scattered.

Solution. The Pharaohs' Twinverse supports heritage conservation by allowing staff to upload artifact media, review damage analysis, record measurements, consult verified references, and prepare cited reports. Conservators can correct results and approve reports at review gates. When evidence is insufficient, the system requests expert examination rather than making unsupported treatment recommendations. Artifact records and decisions are versioned.

Users. Field technicians upload media; conservators review results and approve reports; curators and collection managers maintain artifact records; administrators manage users and permissions; ML engineers and annotators support analysis and evaluation. Heritage authorities oversee data rights. Researchers view artifact, damage and conservation data. Visitors are a future audience.

Why it is architecturally interesting.

- Adaptability. Analysis needs and supported damage categories may evolve; changes should not require redesigning unrelated functions.
- Competing qualities. Responsiveness, service recovery, and low cost must be balanced with expert review, record integrity, and evidence-based reporting.
- Trust built in. The system supports human approval, versioned records, traceable sources, clear uncertainty, and abstention when evidence is weak.
- System responsibilities. Media intake, analysis, review, reports, and records form the core responsibilities.

Scope for this course. The initial prototype covers media intake and quality checks, damage analysis, basic measurements, expert review, cited reports, versioned records, and access control. Image restoration, 3D reconstruction, and the talking statue remain future extensions.

2. Stakeholders and concerns

#	Stakeholder	Role in the project	Key concerns
S1	Conservator	Primary user and final approver at all three review gates. No conservator advisor is identified yet.	Results must be explainable and masks easy to correct. No agreed severity scale. Every claim sourced. Never pushed into a treatment. Professional liability.
S2	Field technician / photographer	Captures and uploads images.	Fast, simple upload. Clear feedback on bad photos, since thin cracks vanish at low resolution. Provenance and licence recorded at upload. Works with unstable connectivity.
S3	Museum curator / collection manager	Business owner.	Artifact safety and ethics. Condition history over time (needs an agreed severity scale). Fit with existing collection records. Cost. Verified facts kept apart from uncertain findings.
S4	Platform administrator / operator	Runs and oversees the platform.	Ease of operation. Recovery from failures. Backups. Cost control (near-zero budget). User management. Cloud use needs permission before heritage images are uploaded.
S5	ML engineer / annotator	Supports analysis, evaluation and annotation.	A fixed artifact-level test split (not by image, to avoid leakage). Complete experiment records. Consistent annotation.
S6	Heritage authority / data-rights holder	Regulator.	Image licences and institutional restrictions: several candidate datasets have blank, unspecified or closed licences, and some model licences restrict regions. No unauthorised data leaving the institution. Audit trail. No unauthorised physical intervention: AI-proposed parts are never printed or attached.
S7	Course team and instructor	Developers and assessor.	Architecture documented with C4/4+1 views and ADRs, and a prototype that conforms to it. Realistic scope: crack detection and segmentation only for the prototype; multi-class damage, restoration, 3D and the talking statue are future extensions. Deadlines.
S8	Researcher	Explores artifacts and reads technical data, with more privileges than a visitor.	Same use cases as a visitor, plus damage measurements and maps, conservation history, technical reports and provenance. AI output and proposals clearly labelled. Data-rights limits respected when exporting reports.
S9	Museum visitor (future)	Future end user.	Not in this release. Uses only the common use cases (browse, 3D, hologram/AR, narration, talking statue), with no technical data. Talking statue and hologram are future extensions.

3. System context (C4 Level 1)

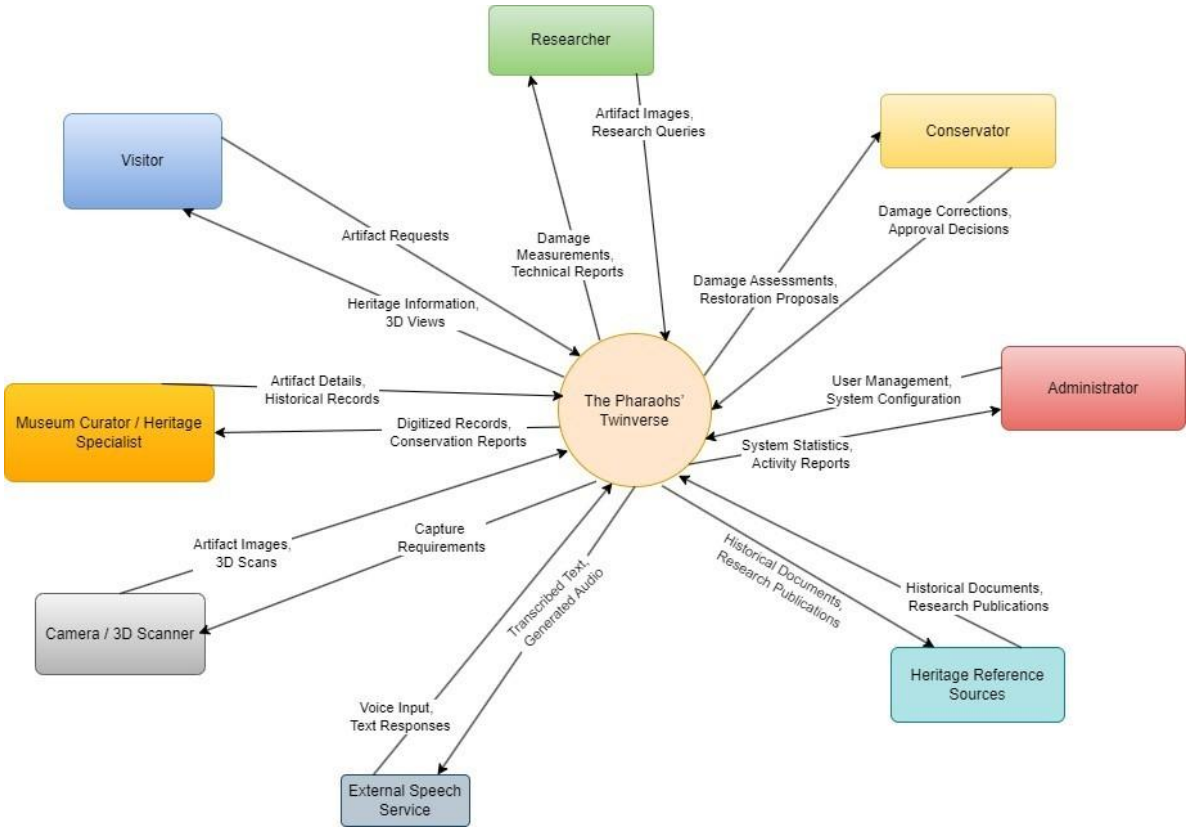


Figure 1. C4 Level 1 context diagram for The Pharaohs' Twinverse.

4. Functional requirements

Requirements are prioritised with MoSCoW. The user stories are adapted from The Pharaohs' Twinverse Project Definition and mapped to the stakeholder roles listed in Section 2. The last column marks what the Milestone 3 prototype will implement.

ID	User story	Role	MoSCoW	In M3 prototype
FR-01	As a field technician/photographer, I want to upload artifact photos, videos, or scans with provenance and licence details and receive media-quality feedback so that I can keep each capture documented and recapture unsuitable media when needed.	Field technician / photographer	Must	Yes
FR-02	As a conservator, I want to confirm the artifact type and material before analysis so that I can ensure the assessment uses the correct context (Gate 1).	Conservator	Must	Yes
FR-03	As a conservator, I want to request damage detection and segmentation and review the marked damage areas so that I can identify where an artifact appears to be damaged.	Conservator	Must	Yes
FR-04	As a conservator, I want to correct and approve damage annotations and retain each accepted edit as a new version so that I can ensure the recorded damage is accurate and traceable (Gate 2).	Conservator	Must	Yes
FR-05	As a conservator, I want to view damage area, affected surface percentage, and crack length and width with clearly labelled units so that I can assess the extent of visible damage.	Conservator	Should	Yes (basic)

FR-06	As a conservator, I want to classify damage using agreed categories and flag uncertain or unclassified cases for review so that I can distinguish identified damage from cases requiring further examination.	Conservator	Should	No (more data)
FR-07	As a conservator, I want to receive an assessment supported by cited conservation sources, or a request for expert examination when evidence is insufficient, so that I can make decisions based on traceable evidence rather than unsupported claims.	Conservator	Must	Yes
FR-08	As a museum curator/collection manager, I want to generate conservation reports that separate verified facts from uncertain findings and show supporting evidence and review status so that I can maintain clear and reliable artifact documentation.	Museum curator / collection manager	Must	Yes
FR-09	As a conservator, I want to approve or reject each report and retain my decision with the report version so that I can ensure a conservation record is not finalized without expert review (Gate 3).	Conservator	Must	Yes
FR-10	As a platform administrator/operator, I want to manage user accounts and role permissions and review audit logs so that I can control access and trace important actions.	Platform administrator / operator	Must	Yes (basic)
FR-11	As an ML engineer/annotator, I want to evaluate candidate damage-analysis approaches on a fixed set of artifacts using agreed quality measures so that I can compare results consistently and support future design decisions.	ML engineer / annotator	Should	Yes (basic)
FR-12	As a museum curator/collection manager, I want to import artifact information from the institution's collection system so that I can keep platform records consistent with existing collection records.	Museum curator / collection manager	Could	No (future)
FR-13	As a conservator, I want to review virtual restoration proposals limited to the approved damaged areas so that I can compare possible visual outcomes without mistaking them for the artifact's confirmed original appearance.	Conservator	Could	No (future)
FR-14	As a conservator, I want to explore a 3D digital representation of an artifact with its damage information so that I can examine the object and the location of damage in three dimensions.	Conservator	Could	No (future)
FR-15	As a museum visitor or researcher, I want to search artifacts, view details and the 3D model, view the statue in hologram or AR, choose a language and listen to narration, and ask the talking statue so that I can explore heritage objects immersively.	Museum visitor (future) / Researcher	Won't (this release)	No
FR-16	As a researcher, I want to view damage measurements and maps, conservation history, technical reports, and artifact provenance and metadata, and optionally export a technical report, so that I can study an artifact's condition and context.	Researcher	Could	No (future)

5. Quality attribute scenarios

Seven scenarios in the six-part form cover seven quality attributes. Every response measure is concrete and testable.

ID	Attribute	Source	Stimulus	Artifact	Environment	Response	Response measure
QAS-1	Performance	Conservator	Submits an artifact image for analysis.	Damage analysis	Normal; 10 concurrent users.	Displays results and progress for longer requests.	Results p95 ≤ 8 s; progress within 1 s; ≥95% of longer requests finish within 60 s.
QAS-2	Availability	Service failure	A task stops during processing.	Analysis and saved records	Active work in progress.	Recovers the task without losing saved work; review remains available.	≥95% of 20 interruption tests recover within 60 s; no uploads or annotations lost; ≥99% of browse/review requests succeed.
QAS-3	Security / integrity	User without approval permission	Attempts to approve a report or change an approved record.	Access and records	Normal operation.	Rejects and audits the action; approved history remains protected.	All 50 tests are rejected and logged within 1 s; all records tested pass integrity checks.
QAS-4	Modifiability	ML engineer / annotator	Adds a damage category or changes an analysis approach.	Analysis capability	Planned development.	Change does not require redesigning review or reporting.	A scoped change takes ≤2 person-days and all regression tests pass.
QAS-5	Usability	First-time conservator	Corrects an annotation and submits it for review.	Review workflow	Desktop browser.	Provides clear correction and undo; saves the accepted edit as a new version.	≥80% of 10 users finish within 5 minutes unaided; usability score ≥70.
QAS-6	Testability	ML engineer / evaluation team	Repeats an evaluation after a change.	Evaluation process	Fixed artifact-level test set available.	Records test data and settings and reports agreed quality measures.	Evaluation takes ≤30 minutes; repeat runs match to 3 decimals; a decline >2 percentage points triggers review.
QAS-7	Trustworthiness	Conservator	Requests an assessment without sufficient evidence.	Assessment and report	Verified conservation references available.	Cites supported claims; abstains and requests expert review when support is weak.	≥95% of 30 unsupported questions lead to abstention; all claims in 50 supported cases are cited; ≥90% of citations pass review.

6. Constraints

6.1 Technical constraints

ID	Constraint
TC-1	The initial prototype focuses on crack detection and segmentation. Supporting additional damage categories requires appropriate labelled data and validation.
TC-2	Candidate analysis approaches must be evaluated using a common, fixed test set.
TC-3	The system must operate within free or low-cost infrastructure and the team's available computing resources. Any paid services require careful budget consideration.
TC-4	The prototype must be demonstrable in an agreed controlled environment.
TC-5	Selected tools, datasets, and services must have licenses suitable for the project.

6.2 Business and domain constraints

ID	Constraint
BC-1	The Pharaohs' Twinverse supports professional decisions and never authorizes physical treatment. Every digital restoration is labelled as a proposal with its evidence, uncertainty, and review status.
BC-2	Heritage images and records may be subject to license and institutional restrictions. They must not be shared with external services without the required permission.
BC-3	Three human review gates are mandatory: confirm the artifact and material, review and correct the damage annotation, and obtain conservator approval before finalizing a record.
BC-4	The conservation reference collection must contain verified, license-checked sources, and source permissions must be recorded.

6.3 Organizational constraints

ID	Constraint
OC-1	No conservator advisor is identified yet. The team should find one to validate the review workflow and the report template.

7. Architecturally significant requirements (ASRs)

7.1 Prioritized requirements

BV = business value, AI = architectural impact (H = high, M = medium, L = low).

Rank	Requirement	BV	AI	ASR
1	Adaptable Damage Analysis: Support changes to damage categories and analysis approaches without redesigning other system functions. Modifiability.	H	H	ASR-1: Adaptable Damage Analysis

Rank	Requirement	BV	AI	ASR
2	Human Review and Versioned Records: Allow conservators to confirm artifact information, correct damage annotations, and approve reports while preserving record history. Security and Integrity.	H	H	ASR-2: Human Review and Versioned Records
3	Evidence-Based Reporting: Generate reports supported by conservation sources and request expert examination when evidence is insufficient. Trustworthiness.	H	H	ASR-3: Evidence-Based Reporting
4	Responsive and Recoverable Operation: Display analysis results promptly and recover interrupted tasks without losing saved data. Performance, Availability, and Low-Cost Infrastructure.	M	H	ASR-4: Responsive and Recoverable Operation
5	Repeatable Analysis Evaluation: Evaluate candidate analysis approaches using a fixed artifact-level test set and consistent quality measures. Testability and Evaluation Baseline.	M	H	ASR-5: Repeatable Analysis Evaluation
6	Heritage Data Protection: Protect restricted heritage images and records from unauthorized sharing. Data Licensing and Institutional Restrictions.	H	M	Addressed by ASR-3 and the context diagram
7	Usable Review Workflow: Make damage annotation correction and report review clear and easy to complete. Usability.	H	M	No
8	Additional Damage Categories: Support more damage categories and flag uncertain cases for further review.	M	M	No (affects ASR-1)
9	Media Intake and Quality Control: Accept artifact photos, videos, and scans with provenance information and quality checks.	H	L	No
10	Damage Measurement and Reporting: Provide damage measurements and reports that distinguish verified facts from uncertain findings.	M	L	No

7.2 Utility tree

Quality attribute	Refinement (scenario)	Scenario	(BV, AI)
Trustworthiness	Cite-or-abstain assessments	QAS-7: Evidence-Based Reporting with Source Citations and Expert Review When Evidence Is Insufficient	(H, H)
	Human approval and versioned records	FR-02/04/09, QAS-3: Artifact and Material Confirmation, Damage Annotation Correction, Report Approval, and Versioned Records	(H, H)
Modifiability	Introduce an analysis capability without redesigning unrelated functions	QAS-4: Add Damage Categories or Change Analysis Approaches Without Redesigning Other Functions	(H, H)
Performance	Display supported results promptly	QAS-1: Display Analysis Results Promptly and Show Progress for Longer Requests	(M, H)
Availability	Recover interrupted tasks without data loss	QAS-2: Recover Interrupted Analysis Tasks Without Losing Saved Data	(M, H)
Testability	Repeatable evaluation of analysis quality	QAS-6: Repeat Analysis Evaluation Using a Fixed Artifact-Level Test Set and Consistent Quality Measures	(M, H)
Security	Role permissions reject unauthorised actions	QAS-3: Enforce Role-Based Access and Record Unauthorized Actions in the Audit Log	(H, M)
Usability	Correct annotation within 5 minutes; usability score ≥70	QAS-5: Enable Conservators to Correct and Submit Damage Annotations Easily, Within Five Minutes	(H, M)

7.3 Why the top five shape the architecture

ASR-1: Adaptable damage-analysis capability (modifiability)

- Why: Damage categories and analysis approaches may evolve as evidence grows. The system should support change without redesigning unrelated functions.
- Architectural implication: Keep analysis responsibilities distinct from review, reporting, and record management.

ASR-2: Human review gates and versioned records (trustworthiness, security)

- Why: Incorrect results can mislead conservation decisions, so expert review and traceability are mandatory.
- Architectural implication: Enforce the three review gates, role permissions, version history, and audit records.

ASR-3: Evidence-based reporting with abstention (trustworthiness)

- Why: Unsupported conservation claims can mislead users. Reports must cite sources, separate facts from uncertainty, and avoid unsupported advice.
- Architectural implication: Evidence retrieval and reporting must provide traceable sources and abstain when reliable support is missing.

ASR-4: Responsive and recoverable operation within budget (performance, availability, cost)

- Why: Users need timely feedback, but analysis takes time and resources. Failures must not lose uploads or approved work.

- Architectural implication: Show progress, recover interrupted work, and keep saved records accessible.
- Trade-off: Recovery features increase complexity, while limited resources restrict concurrent work.

ASR-5: Repeatable evaluation of analysis quality (testability)

- Why: The team needs comparable evidence to evaluate candidate analysis approaches and justify design choices.
- Architectural implication: Use a fixed artifact-level test set, documented criteria, and recorded results.

7.4 Competing quality attributes

Quality attributes may compete.

- Performance vs trustworthiness: evidence checks can increase response time, but reports must remain trustworthy.
- Security and integrity vs usability: review gates and protected records add steps for users.
- Modifiability vs performance: flexible analysis capabilities must be balanced against timely results.

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